

ESP8266

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ESP8266



Participants	
Skills	Electronics, Software
Status	Getting parts
Niche	Electronics
Purpose	Education
Tool	No
Location	
Cost	about 5 dollar per module
Tool category	

Current status

- ▶ Modules ordered awaiting shipment

links

- ▶ SDK documentation (all chinees) DOCS (<http://rghost.net/57965174%7C>)
- ▶ VM file [1] (<http://pan.baidu.com/s/1dDf8Iy9>) Password: i90l
- ▶ Form about the module <http://www.esp8266.com/>
- ▶ seeedstudio.com/depot/WiFi-Serial-Transceiver-Module-w-ESP8266-p-1994.html
- ▶ ESP8266 ROM Bootloader utility <https://github.com/themadinventor/esptool>

Datasheet

- ▶ English Datasheet
- ▶ [http://www.seeedstudio.com/document/pdf/ESP8266%20Specifications\(Chinese\).pdf](http://www.seeedstudio.com/document/pdf/ESP8266%20Specifications(Chinese).pdf) (Chinese)

Introduction

Yue Xin intelligent high performance wireless connectivity platform --ESCP SOC, designers bring the Gospel to the mobile platform, it At the lowest cost to provid maximum usability for WiFi capabilities embedded in other systems offer unlimited possibilities.

Technical Overview

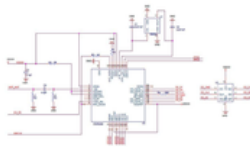
ESP8266 is a complete and self-contained Wi-Fi network solutions that can carry software applications, or through Another application processor uninstall all Wi-I networking capabilities. ESP8266 when the device is mounted and as the only application of the application processor, the flash memory can be started directly from an external Move. Built-in cache memory will help improve system performance and reduce memory requirements. Another situation is when wireless Internet access assume the task of Wi-Fi adapter, you can add it to any microcontroller-based design, the connection is simple, just by SPI / SDIO interface or central processor AHB bridge interface. Processing and storage capacity on ESP8266 powerful piece, it can be integrated via GPIO ports sensors and other

applications specific equipment to achieve the lowest early in the development and operation of at least occupy system resources. The ESP8266 highly integrated chip, including antenna switch balun, power management converter, so with minimal external circuitry, and includes front-end module, including the entire solution designed to minimize the space occupied by PCB. The system is equipped with ESP8266 manifested leading features are: energy saving VoIP quickly switch between the sleep / wake patterns, with low-power operation adaptive radio bias, front-end signal processing functions, troubleshooting and radio system coexist characteristics eliminate cellular / Bluetooth / DDR / LVDS / LCD interference.

Characteristics

- ▶ 802.11 b / g / n
- ▶ Wi-Fi Direct (P2P), soft-AP
- ▶ Built-in TCP / IP protocol stack
- ▶ Built-in TR switch, balun, LNA, power amplifier and matching network
- ▶ Built-in PLL, voltage regulator and power management components
- ▶ 802.11b mode + 19.5dBm output power
- ▶ Built-in temperature sensor
- ▶ Support antenna diversity
- ▶ off leakage current is less than 10uA
- ▶ Built-in low-power 32-bit CPU: can double as an application processor
- ▶ SDIO 2.0, SPI, UART
- ▶ STBC, 1x1 MIMO, 2x1 MIMO
- ▶ A-MPDU, A-MSDU aggregation and the 0.4 Within wake
- ▶ 2ms, connect and transfer data packets
- ▶ standby power consumption of less than 1.0mW (DTIM3)

Schema



Ultra-low power technology

ESP8266 specifically for mobile devices, wearable electronics and networking applications design and make the machine to achieve the lowest energy consumption, together with several other patented technology. This energy-efficient construction in three modes: active mode, sleep mode and deep sleep mode type. When ESP8266 using high-end power management technology and logic systems to reduce non-essential functions of the power conversion regulate sleep patterns and work modes, in sleep mode, it consumes less than the current 12uA, is connected, it consumes less power to 1.0mW (DTIM = 3) or 0.5mW (DTIM = 10). Sleep mode, only calibrated real-time clock and watchdog in working condition. Real-time clock can be programmed to wake ESP8266 within a specific period of time. Through programming, ESP8266 will automatically wake up when detected certain to happen. ESP8266 automatic wake-up in the shortest time, this feature can be applied to the SOC for mobile devices, so before you turn Wi-Fi SOC are in a low-power standby mode. To meet the power requirements of mobile devices and wearable electronics products, ESP8266 at close range when the PA output power can be reduced through software programming to reduce overall power consumption in order to adapt to different applications.

Maximum integration

ESP8266 integrates the most critical components on the board, including power management components, TR switch, RF balun, a peak power of + 25dBm of PA, therefore, ESP8266 only guarantee the lowest BOM cost, and easy to be embedded in any system. ESP8266 BOM is the only external resistors, capacitors, and crystal.

ESP8266 application subject

- ▶ Smart Power Plug
- ▶ Home Automation
- ▶ mesh network
- ▶ industrial wireless control
- ▶ Baby Monitor
- ▶ Network Camera
- ▶ sensor networks
- ▶ wearable electronics
- ▶ wireless location-aware devices
- ▶ Security ID tag
- ▶ wireless positioning system signals

Specifications

Power

The following data are based on a 3.3V power supply, ambient temperature 25C and use the internal regulator measured. [1] All measurements are made in the absence of the SAW filter, the antenna interface is completed. [2] all transmit data based on 90% duty cycle, continuous transmission mode in the measured.

Mode	Min	Typical	Max	Units
802.11b, CCK 1Mbps, POUT=+19.5dBm		215		mA
802.11b, CCK 11Mbps, POUT=+18.5dBm		197		mA
802.11g, OFDM 54Mbps, POUT=+16dBm		145		mA
802.11n, MCS7, POUT =+14dBm		135		mA
802.11b, packet size of 1024 bytes, -80dBm		60		mA
802.11b, packet size of 1024 bytes, -70dBm		60		mA
802.11b, packet size of 1024 bytes, -65dBm		62		mA
Standby		0.9		mA
Deep sleep		10		mA
Saving mode DTIM 1		1.2		mA
Saving mode DTIM 3		0.86		mA
Shutdown		0.5		mA

RF specifications

The following data is at room temperature, the voltage of 3.3V and 1.1V, respectively, when measured

Description	Min	Typical	Max	Units
Input Frequency	2412		2484	MHz
Input resistance		50		Ω
Input reflection			-10	dB
At 72.2Mbps, PA output power	14	15	16	dBm
11b mode, PA output power	17.5	18.5	19.5	dBm
Sensitivity				
CCK, 1Mbps		-98		dBm
CCK, 11Mbps		-91		dBm
6Mbps (1/2 BPSK)		-93		dBm
54Mbps (3/4 64-QAM)		-75		dBm
HT20, MCS7 (65Mbps, 72.2Mbps)		-71		dBm
Adjacent suppression				
OFDM, 6Mbps		37		dB
OFDM, 54Mbps		21		dB
HT20, MCS0		37		dB
HT20, MCS7		20		dB

CPU and memory

CPU Interface

The chip embedded in an ultra-low-power 32-bit micro-CPU, with 16 compact mode. Can be connected to the CPU via the following interfaces:

- ▶ connecting storage controllers can also be used to access external code memory RAM / ROM interface (iBus)
- ▶ Also attached storage controller data RAM interface (dBus)
- ▶ Access Register of AHB interface
- ▶ JTAG debug interface

Storage Controller

Storage controller contains ROM and SRAM. CPU can iBus, dBus and AHB interface to access the storage controller. Any one of these interfaces can apply for access to ROM or RAM cells, memory arbiter to determine the running order in the order of arrival.

AHB and AHB module

AHB module acts as arbiter, through the MAC, and SDIO host CPU control AHB interface. Since sending Address different, AHB data requests may arrive the following two slaves in one: APB module, or flash memory controller (usually in the case of off-line applications) to the received request is a high speed memory controllers often request, APB module receives register access is often Request. APB module acts as a decoder, but only you can access the ESP8266 main modu programmable registers. Since the sending address different, APB request may reach the radio receiver, SI / SPI, hosts SDIO, GPIO, UART, real-time clock (RTC, MAC or digital baseband.

Interface

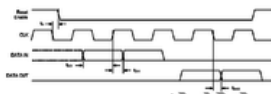
ESP 8266 contains multiple analog and digital interfaces, as follows:

Main SI / SPI control (optional)

Main Serial Interface (SI) can run at two, three, four-wire bus configuration, is used to control the EEPROM or other I2C / SPI devices. Multiple devices share the two-wire I2C bus. Multiple SPI devices to share the clock and data signals, and according to the chip select, each controlled by software alone GPIO pins. SPI can be used to control external devices, such as serial flash, audio CODEC or other slave devices, installation, effectively giving it three different pins, making it the standard master SPI device.

- ▶ SPI_EN0
- ▶ SPI_EN1
- ▶ SPI_EN2

SPI slave is used as the primary interface, giving SPI master and slave SPI support. In the built-in applications, SPI_EN0 is used as an enable signal, the role of external serial flash, download firmware and / or MIB data to baseband. In host-based applications, the firmware and you can choose one MIB data downloaded via the host interface both. This pin is active low when not should be left unconnected. SPI_EN1 often used for user applications, such as controlling the built-in applications or external audio codec sensor ADC. This pin is active low when not should be left unconnected. SPI_EN2 often used to control the EEPROM, storing individual data (individual data), such as MIB information, MAC address, and calibration data, or for general purposes. This pin is active low when not should be left unconnected.



General Purpose IO

A total of up to 16 GPIO pins. The firmware can assign them different functions. Each GPIO can be configured internal pullup / pulldown resistors available software registers sampled input, triggering edge or level CPU interrupt input, trigger level wake-up interrupt input, open-drain or complementary push-pull output drivers, software register output source or sigma-delta PWM DAC. These pins are multiplexed with other functions, such as the main interface, UART, SI, Bluetooth co-existence and so on.

Digital IO pins

Digital IO pad is two-way, three states. It includes a three-state control input and output buffers. In addition, for low-power operation, IO can be set to hold state. For example, when we reduce the chip's power consumption, all the output enable signal can be set to maintain a low-power state. Hold function can be selectively implanted IO in need. When the IO help internal and external circuit driving, hold function can be used to hold last state. Hold function to pin introduce some positive feedback. Therefore, the external drive pin must be stronger than the positive feedback. However, the required driving force size is still small, in the 5uA of.

Variables	Symbol	Min	Max	Units
Input Low Voltage	Vil	-0.3	0.25xV10	V
Input High Voltage	Vih	0.75xV10	3.6	V
Input leakage current	IIL	-	50	nA
Output Low Voltage	VOL	-	0.1xV10	V
Output High Voltage	VOH	0.8xV10	-	V
Input pin capacitance	Cpad	-	2	pF
VDDIO	V10	1.7	3.6	V
Current	Imax	-	12	mA
Temperature	Tamb	-20	100	C

All digital IO pins must add an overvoltage protection circuit (snap back circuit) between the pin and ground. Usually bounce (snap back) voltage is about 6V, while maintaining the voltage is 5.8V. This prevents excessive voltage and generating ESD. Diodes also avoid reverse voltage output devices.

Firmware and software tools development kit

The firmware running on the ROM and SRAM chip, when the device is awake, firmware via SDIO sector Download the instructions from the host side. Firmware is fully compliant with 802.11 b / g / n / e / i WLAN MAC protocol and Wi-Fi Direct specification only supports basic services unit distributed control function (DCF) under (BSS) operation, but also follow the latest Wi-Fi P2P protocol to support P2P groups operating (P2P group operation). Low level protocol functions automatically run by ESP8266, such as

- ▶ RTS / CTS
- ▶ Confirm
- ▶ fragmentation and reassembly
- ▶ polymerization
- ▶ frame package (802.11h / RFC 1042)
- ▶ automatic beacon monitoring / scanning
- ▶ P2P WiFi direct

With P2P discovery procedures, passive or active scanning once in the host command start, it will be done automatically. Perform power management, interaction with the host at least, this way, the task of effectively minimized.

Features

Laboratory features of the software developer's kit is as follows:

- ▶ 802.11 b / g / n / d / e / i / k / r support
- ▶ Wi-Fi Direct (P2P) support
- ▶ P2P discovery, P2P group master mode (Group Owner mode), P2P Power Management
- ▶ Infrastructure Network (Infrastructure BSS) station (Station) mode / P2P mode / SoftAP mode
- ▶ hardware accelerator
 - ▶ CCMP (CBC-MAC, counting mode)
 - ▶ TKIP (MIC, RC4) o WAPI (SMS4)
 - ▶ WEP (RC4)
 - ▶ CRC
 - ▶ WPA / WPA2 and WPS support
 - ▶ Other 802.11i security features:
- ▶ Pre-Certification
- ▶ TSN

Open interfaces □ prepared for a variety of upper EAP authentication methods, such as:

- ▶ TLS
- ▶ PEAP
- ▶ LEAP
- ▶ SIM
- ▶ AKA
- ▶ 802.11n support (2.4GHz / 5GHz)
- ▶ Support MIMO 1x1 and 2x1, STBC, A-MPDU and A-MSDU aggregation, 0.4s guard interval
- ▶ WMM saving U-APSD

- ▶ use with multi-queue QoS management, in line with 802.11e standard multimedia data traffic optimization methods
- ▶ Follow the UMA, and certified by UMA
- ▶ 802.11h / RFC1042 frame encapsulation
- ▶ hash DMA data transfer operation, the CPU usage to a minimum
- ▶ antenna diversity and choice (by software management hardware)
- ▶ the clock / power gating and follow the 802.11 standard power management combined, according to the current connection, enter OK dynamically adjusted to achieve the lowest energy consumption
- ▶ ratio can be adjusted to set an optimum algorithm for the missing data and the Tx power transfer rate based on the actual SNR and packet

Rate

- ▶ MAC layer automatic retransmission and automatic response, to avoid packet loss occurs when the host is running slow
- ▶ seamless roaming support
- ▶ Configurable packet traffic arbitration and tailored, based on the slave processor design combining a series of Bluetooth chip vendors to provide flexible and precise time-Bluetooth coexistence support
- ▶ support dual / single antenna Bluetooth coexistence with syncing WiFi / Bluetooth capability

Power Management

Chip can tune into the following states:

- ▶ off (OFF): CHIP_PD pin is in a low power state. RTC failure. All registers are emptied.
- ▶ deep sleep (DEEP_SLEEP): RTC open, other parts of the chip are closed. RTC internal recovery memory to save the basic WiFi connection information.
- ▶ sleep (SLEEP): Only RTC running. Crystal oscillator stops. Any part of the wake (MAC, host, RTC timer, external interrupt) will make the wake of the chip.
- ▶ Wake (WAKEUP): In this state, the system from a sleep state to start (PWR) status. Crystal oscillator and PLL are converted enabled state.

* on state (ON): High-speed clock can run, And sent to each clock control register is enabled Modules. Each module, including the CPU, including the implementation of relatively low-level clock gating. When the system works, you can WAITI instructions to turn off the CPU's internal clock.



Clock Management

High Frequency Clock

ESP8266 on high frequency clock is used to drive two Tx and Rx mixer, which is generated by the internal oscillator and an external oscillator. Crystal frequency between 26MHz to 52MHz float. Although the internal crystal oscillator of the calibration range of the crystal so that the clock generator to meet the conditions, but in general, the crystal quality is still obtained a proper phase noise factors to be considered. When the crystal is used, or because of the frequency offset, rather than the best choice for quality, the maximum capacity of the data processing system and will reduce the sensitivity of the wifi. Please refer to the follow instructions to measure the frequency offset.

Variables	Symbol	Min	Max	Units
Frequency	Fxo	52		MHz
Load capacitance	Cl		32	pF
Dynamic capacitance	Cm	2	5	pF
Serial resistance	Rs	0	65	Ω
Frequency tolerance	Fxo	-15	15	ppm
Frequency vs Temperature (-25C ~ 75C)	Fxo,Temp	-15	15	ppm

External Reference Requirements

At 26MHz external clock frequency between 52MHz. In order to make a well-functioning radio receiver, the clock will Must have the following characteristics:

Variables	Symbol	Min	Max	Units
Clock amplitude	Vxo	0.2	1	Vpp
External clock accuracy	Fxo,EXT	-15	15	ppm
Phase Noise @ 1kHz offset, 40MHz clock			-120	dBc/Hz
Phase Noise @ 10kHz offset, 40MHz clock			-130	dBc/Hz
Phase Noise @ 100kHz offset, 40MHz clock			-138	dBc/Hz

Radio receivers

ESP8266 radio receiver mainly includes the following modules:

- ▶ 2.4GHz receiver
- ▶ 2.4GHz transmitter
- ▶ High-speed clock generator and crystal oscillator
- ▶ Real-time clock
- ▶ bias and regulators
- ▶ Power Management

Channel Frequency

According IEEE802.11bgn standard, RF transceiver supports the following channels:

Channel	Freq.	Channel	Freq.
1	2412	8	2447
2	2417	9	2452
3	2422	10	2457
4	2427	11	2462
5	2432	12	2467
6	2437	13	2472
7	2442	14	2484

2.4GHz receiver

2.4GHz RF signal receiver down into quadrature baseband signal, with two high-resolution, high-speed ADC and the latter into a digital signal. In order to accommodate different signal channels, a radio receiver integrated RF filters, automatic gain control (AGC), DC offset compensation circuit and a baseband filter.

2.4GHz transmitter

2.4 GHz transmitter orthogonal frequency baseband signals up to 2.4GHz, using high-power CMOS power amplifier to drive the antenna. Further use of the digital calibration improves the linearity of the power amplifier to achieve the average power of + 19dBm in 802.11b transmission, the transmission reaches + 16dBm 802.11n average power, features super. To offset defects in the radio receiver is also calibrated by other measures such as:

- ▶ carrier leakage
- ▶ I / Q phase matching, and
- ▶ baseband nonlinear

This will reduce the time and equipment required for testing.

Clock generator

The clock generator generates the receiver and transmitter 2.4GHz clock signal all of its components are integrated on the chip, Include:

- ▶ inductor
- ▶ varactor
- ▶ closed-loop filter

Clock generator contains a built-in calibration circuit and self-test circuitry. Clock phase and quadrature phase noise through the optimal calibration algorithm processing patent on the chip, in order to ensure that the receiver and transmitter to achieve the best performance.

AT Commands

- ▶ Tnx to <http://www.electrodragon.com/w/Wi07c>

Format

- ▶ Baud rate at 57600
- ▶ x is the commands

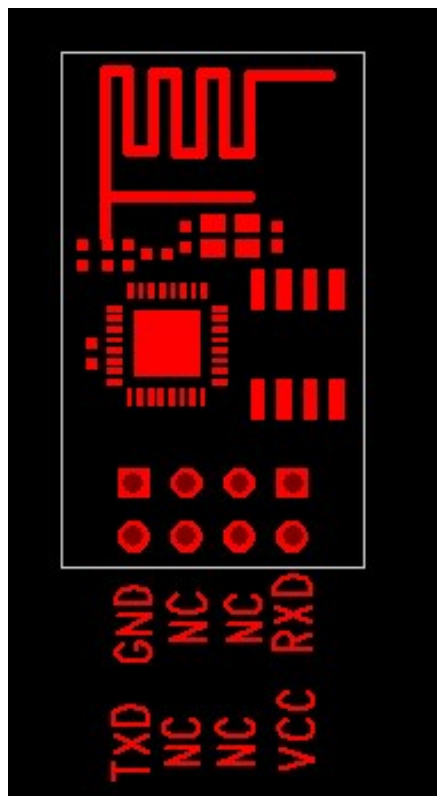
Set	Inquiry	Test	Execute
AT+<x>=<...>	AT+<x>?	AT+<x>=?	AT+<x>
AT+CWMODE=<mode>	AT+CWMODE?	AT+CWMODE=?	-
Set the network mode	Check current mode	Return which modes supported	-

Commands

- ▶ carefully there are must be no any spaces between the " and IP address or port

Commands	Description	Type	Set/Execute	Inquiry	test	Parameters	Examples
AT+RST	restart the module	basic	-	-	-	-	
AT+CWMODE	wifi mode	wifi	AT+CWMODE=<mode>	AT+CWMODE?	AT+CWMODE=?	1= Sta, 2= AP, 3=both	
AT+CWJAP	join the AP	wifi	AT+ CWJAP = <ssid>,<pwd>	AT+ CWJAP?	-	ssid = ssid, pwd = wifi password	
AT+CWLAP	list the AP	wifi	AT+CWLAP				
AT+CWQAP	quit the AP	wifi	AT+CWQAP	-	AT+CWQAP=?		
AT+ CWSAP	set the parameters of AP	wifi	AT+ CWSAP= <ssid>,<pwd>,<chl>,<ecn>	AT+ CWSAP?		ssid, pwd, chl = channel, ecn = encryption	Connect to your router: ; AT+CWJAP="YOURSSID","helloworld"; and check if connected: AT+CWJAP?
AT+ CIPSTATUS	get the connection status	TCP/IP	AT+ CIPSTATUS				
AT+CIPSTART	set up TCP or UDP connection	TCP/IP	1)single connection (+CIPMUX=0) AT+CIPSTART= <type>,<addr>,<port>; 2) multiple connection (+CIPMUX=1) AT+CIPSTART= <id>,<type>,<addr>,<port>	-	AT+CIPSTART=?	id = 0-4, type = TCP/UDP, addr = IP address, port= port	Connect to another TCP server, set multipl connection first: AT+CIPMUX=1; connect: AT+CIPSTART=4,"TCP","X1.X2.X3.X4",995
AT+CIPSEND	send data	TCP/IP	1)single connection(+CIPMUX=0) AT+CIPSEND= <length>; 2) multiple connection (+CIPMUX=1) AT+CIPSEND= <id>,<length>		AT+CIPSEND=?		send data: AT+CIPSEND=4,15 and then enter the data
AT+CIPCLOSE	close TCP or UDP connection	TCP/IP	AT+CIPCLOSE=<id> or AT+CIPCLOSE		AT+CIPCLOSE=?		
AT+CIFSR	Get IP address	TCP/IP	AT+CIFSR		AT+ CIFSR=?		
AT+ CIPMUX	set mutiple connection	TCP/IP	AT+ CIPMUX=<mode>	AT+ CIPMUX?		0 for single connection 1 for mutiple connection	
AT+ CIPSERVER	set as server	TCP/IP	AT+ CIPSERVER= <mode>[,<port>]			mode 0 to close server mode, mode 1 to open; port = port	turn on as a TCP server: AT+CIPSERVER=1,8888, check the self server IP address: AT+CIFSR=?
+IPD	received data						

Pin Definition



Category: Projects

